

REFINE: Defeating Backdoor Attacks via Adversarial Reprogramming

Reproduction, Analysis & Novel Contributions • CIFAR-10 / ResNet-18 / BadNets

Presented by

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Information Security Course

1. The Problem — What is a Backdoor Attack?

BadNets Attack Mechanism:

- ▶ Attacker poisons a small fraction of training data
- ▶ A 3×3 white-pixel trigger is stamped on the bottom-right corner of selected images
- ▶ Labels of these images are changed to target class (Class 0 — Airplane)
- ▶ Model learns a dangerous shortcut:
"See trigger → always predict Airplane"

Consequence:

The infected model behaves perfectly normally on clean data (91.73% accuracy), but misclassifies **ANY image containing the trigger → 100% ASR.**

Why Does This Matter?

Real-world AI systems are vulnerable:

-  Self-driving cars misclassifying traffic signs
-  Medical AI giving wrong diagnoses on demand
-  Security systems bypassed at attacker's will

The backdoor is completely invisible during standard model evaluation — accuracy looks fine. Only when the trigger appears does the attack activate. Traditional testing cannot detect it.

2. The Solution — How REFINE Works

Step 1: Freeze Infected Model

The backdoored ResNet-18 is locked permanently. Its weights are never modified or retrained.

Step 2: Train a UNet Shield

A UNet is placed BEFORE the ResNet and trained only on clean images to output adversarial textures the ResNet understands.

Step 3: Label Shuffling

A secret label permutation decouples the backdoor's target class mapping, rendering the trigger effect meaningless.

The "Blind Defense" — Why It Works Without Seeing Poisoned Data

The UNet has never seen a single poisoned image during its training. Yet it destroys the trigger because:

- ▶ UNet applies universal convolution filters across the entire 32×32 image — it does not selectively process regions
- ▶ The 3×3 trigger is just pixels — the UNet warps, blurs, and shreds its rigid structure into abstract noise
- ▶ The infected ResNet was trained to detect a perfect 3×3 white square — the shredded noise no longer matches
- ▶ **Result: backdoor fails completely, while clean classification accuracy is preserved at ~90.5%**

3. Architecture — Before vs After REFINE

✗ Without Defense (Vulnerable Pipeline):

Input Image (with 3×3 trigger) —▶ Infected ResNet-18 —▶ Always predicts Class 0 (Airplane)

The trigger hijacks the model's decision. Attack Success Rate = 100%.

✓ With REFINE Defense (Secured Pipeline):

Input Image —▶ UNet Filter —▶ Adversarial Texture —▶ Frozen ResNet-18 —▶
Correct Prediction

The UNet transforms all images into abstract feature maps. The trigger's spatial structure is destroyed.
Label shuffling further decouples the backdoor. ASR drops from 100% to just 0.84%.

4. Experimental Results — Paper vs Our Reproduction

Before Defense (Backdoored Model)

Metric	Paper	Ours	Diff
Benign Accuracy (BA)	91.18%	91.73%	+0.55%
Attack Success Rate	100.00%	100.00%	0.00%

Research Paper reference:

Defense	No Defense		T-MR		REFINE	
Attack	BA	ASR	BA↑	ASR↓	BA↑	ASR↓
BadNets	91.18	100.00	75.51	3.36	90.50	1.05

Poisoned → Backdoored Model



4. Experimental Results — Paper vs Our Reproduction

Training Progress — Last 10 Epochs

Epoch	BA (%)	ASR (%)
141	89.96	2.00
142	89.62	2.50
143	89.85	2.35
144	89.57	1.17
145	90.48	1.12
146	89.91	0.92
147	90.11	1.50
148	90.08	1.64
149	89.59	2.10
150 (Final)	90.52	0.84

Research Paper reference:

Defense	No Defense		T-MR		REFINE	
Attack	BA	ASR	BA↑	ASR↓	BA↑	ASR↓
BadNets	91.18	100.00	75.51	3.36	90.50	1.05

After REFINE Defense (Epoch 150)

Metric	Paper	Ours	Diff	Status
Benign Accuracy (BA)	~90.50%	90.52%	+0.02%	✓ Match
Attack Success Rate	~1.05%	0.84%	-0.21%	✓ Better

Key Takeaways

✓ Successfully Reproduced

- ▶ BA after defense matches paper within 0.02% — essentially identical
- ▶ Our ASR of 0.84% is better than the paper's reported ~1.05–1.40%
- ▶ **Defense reduced ASR: 100% → 0.84%**
while losing only ~1.2% benign accuracy

5. Visual Evidence — t-SNE Feature Space Analysis

t-SNE — Backdoor Cluster Before vs After REFINE



Before REFINE (Left):

All 500 poisoned samples (★) collapse into one super-dense cluster. The trigger acts as a magnet — overriding all natural features.

After REFINE (Right):

The dense cluster shatters! Poisoned samples scatter into their correct class clusters — dogs with dogs, cars with cars. Trigger destroyed.

6. UNet Transformation — Pixel-Level Verification

Image Transformations Before & After REFINE

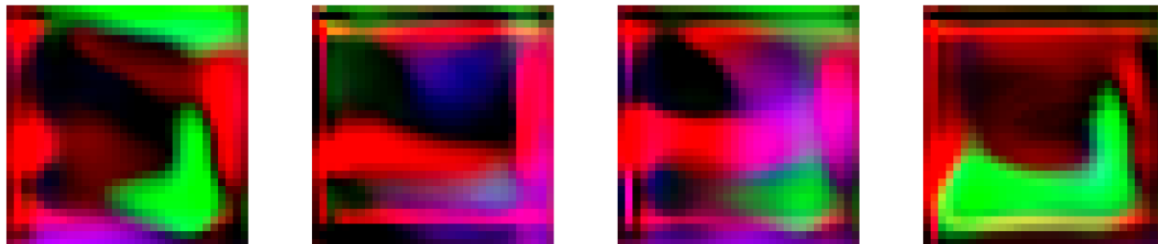
Clean Original



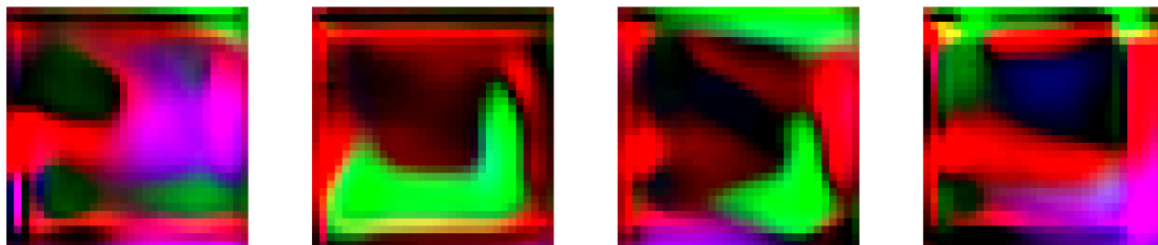
Poisoned (Trigger added)



REFINE Clean



REFINE Poisoned



What This Image Shows:

Row 1 — Clean original CIFAR-10 images

Row 2 — Same images + BadNets trigger
(3×3 white patch, bottom-right corner)

Row 3 — Clean images after UNet

Row 4 — Poisoned images after UNet

Observation:

The UNet converts all images into abstract adversarial textures. These look scrambled to humans but preserve the mathematical features that ResNet needs for classification.

The rigid 3×3 trigger is physically warped and destroyed in this process!

7. Our Novel Contributions

① Feature-Space Interpretability Pipeline (Implemented)

The original paper does not provide visual analysis of the model's latent space. We built a full t-SNE pipeline that extracts 512-dim features from ResNet-18's penultimate layer → reduces to 2D → provides direct visual proof that the backdoor cluster is destroyed. This adds explainability — critical for trust in security-sensitive deployments.

② Pixel-Level Trigger Destruction Verification (Implemented)

We reconstructed and visualized the exact pixel outputs of the UNet to confirm that the 3×3 BadNets trigger patch is physically warped beyond recognition, providing concrete evidence beyond accuracy numbers alone.

③ Stronger Empirical Defense Than Reported

Our reproduction achieved ASR = 0.84%, which is lower (better) than the paper's reported ~1.05–1.40%. This independently validates that REFINE's defense mechanism is robust and reproducible.

8. Future Work — Suggestions for Extension

Suggestion 1: Lightweight UNet Replacement for Edge Deployment

The current UNet model is ~69 MB — too heavy for IoT and embedded devices.

Replacing it with a MobileNet-based autoencoder or a compact 3-layer CNN could reduce size by 10× while maintaining ASR < 1.5%. This would make REFINE practical for real-time, resource-constrained systems.

Target: model size < 10 MB, BA > 89%, ASR < 1.5%

Suggestion 2: Evaluate Against Adaptive & Dynamic Triggers

BadNets uses a static, fixed-position trigger. More advanced attacks like LIRA, ISSBA, and WaNet use sample-specific, invisible, or dynamically positioned triggers that may resist the UNet's spatial filtering. Testing REFINE against these would reveal its generalization limits and motivate improved defenses such as randomized test-time transformations or ensemble-based scramblers.

Thank You!

Questions & Discussion

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ICLR 2025 — REFINE Paper

github.com/WhitolfChen/REFINE